



**UNIVERSIDAD PERUANA
CAYETANO HEREDIA**

FACULTAD DE CIENCIAS E INGENIERÍA

CURSO:

SISTEMAS OPERATIVOS

TRABAJO ACADÉMICO 2

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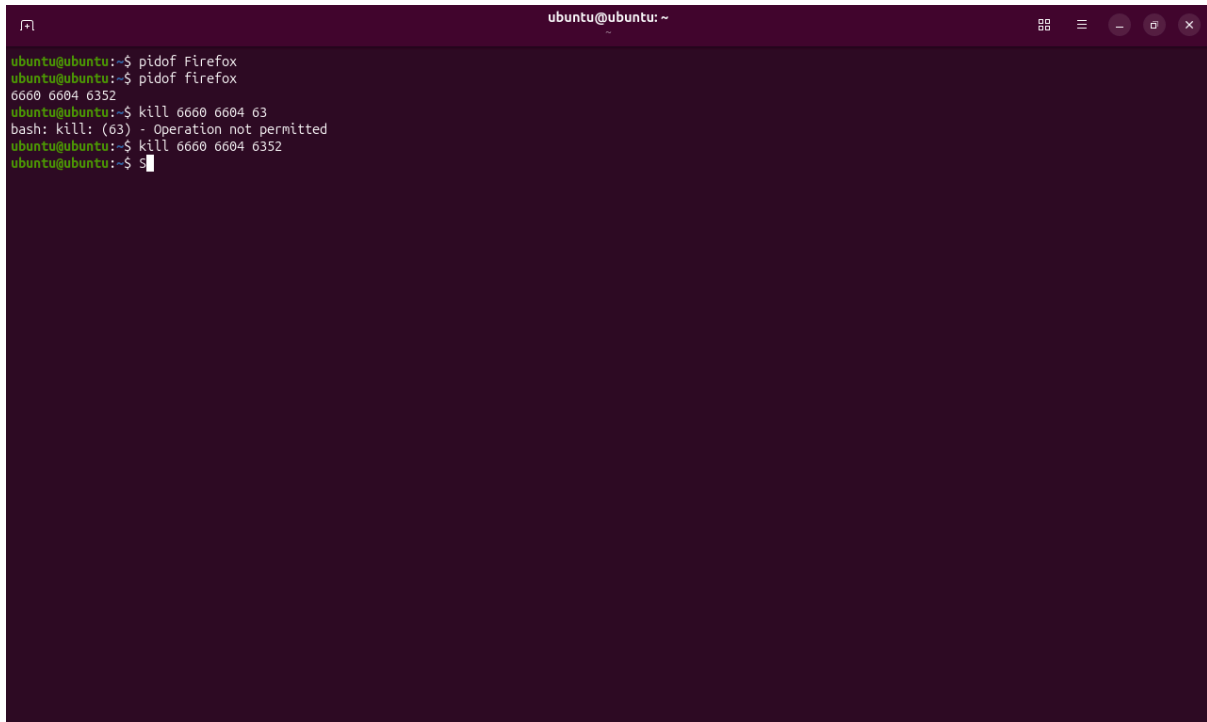
2026

ACTIVIDAD

Revisar el material de Comandos Linux de esta sesión respecto a los comandos que tiene que ver con las actividades de los procesos.

Ejemplo:

• Kill

A terminal window titled 'ubuntu@ubuntu: ~' with standard window controls. The terminal shows a sequence of commands and their outputs: 'pidof Firefox' returns '6660 6604 6352'; 'pidof firefox' returns the same PIDs; 'kill 6660 6604 63' results in an error 'bash: kill: (63) - Operation not permitted'; and 'kill 6660 6604 6352' is entered without output. The prompt is always 'ubuntu@ubuntu:~\$'.

```
ubuntu@ubuntu:~$ pidof Firefox
ubuntu@ubuntu:~$ pidof firefox
6660 6604 6352
ubuntu@ubuntu:~$ kill 6660 6604 63
bash: kill: (63) - Operation not permitted
ubuntu@ubuntu:~$ kill 6660 6604 6352
ubuntu@ubuntu:~$
```

El comando kill es una herramienta fundamental de la interfaz de línea de comandos (CLI) en sistemas operativos tipo Unix, diseñada para enviar señales específicas a los procesos en ejecución. Aunque su nombre sugiere la destrucción del proceso, su función real es la comunicación entre procesos mediante señales del sistema, siendo la terminación de programas su aplicación más común.

• Killall

```
ubuntu@ubuntu: ~  
ubuntu@ubuntu:~$ killall  
Usage: killall [OPTION]... [--] NAME...  
       killall -l, --list  
       killall -V, --version  
  
-e,--exact          require exact match for very long names  
-I,--ignore-case    case insensitive process name match  
-g,--process-group  kill process group instead of process  
-y,--younger-than   kill processes younger than TIME  
-o,--older-than     kill processes older than TIME  
-i,--interactive    ask for confirmation before killing  
-l,--list           list all known signal names  
-q,--quiet          don't print complaints  
-r,--regex          interpret NAME as an extended regular expression  
-s,--signal SIGNAL  send this signal instead of SIGTERM  
-u,--user USER      kill only process(es) running as USER  
-v,--verbose        report if the signal was successfully sent  
-V,--version        display version information  
-w,--wait           wait for processes to die  
-n,--ns PID         match processes that belong to the same namespaces  
                    as PID  
-Z,--context REGEXP kill only process(es) having context  
                    (must precede other arguments)  
  
ubuntu@ubuntu:~$ killall -l  
HUP INT QUIT ILL TRAP ABRT BUS FPE KILL USR1 SEGV USR2 PIPE ALRM TERM STKFLT  
CHLD CONT STOP TSTP TTIN TTOU URG XCPU XFSZ VTALRM PROF WINCH POLL PWR SYS  
ubuntu@ubuntu:~$ killall firefox  
ubuntu@ubuntu:~$
```

El comando `killall` es una utilidad de administración de procesos que permite enviar señales a los procesos basándose en su nombre, en lugar de su identificador numérico (PID). A diferencia de `kill`, que actúa sobre una única instancia específica, `killall` afecta de manera simultánea a todas las instancias activas que coincidan exactamente con el nombre del comando de ejecución especificado.

• Top

```

ubuntu@ubuntu: ~ -- top
top - 19:31:00 up 55 min, 1 user, load average: 0.45, 0.51, 0.56
Tasks: 364 total, 2 running, 361 sleeping, 0 stopped, 1 zombie
%Cpu(s): 0.0 us, 4.4 sy, 0.0 ni, 93.8 id, 0.0 wa, 0.0 hi, 1.8 si, 0.0 st
MiB Mem : 15986.0 total, 2615.0 free, 3697.0 used, 10677.1 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used, 12289.0 avail Mem

  PID USER      PR  NI   VIRT    RES    SHR  S  %CPU  %MEM    TIME+  COMMAND
 3863 ubuntu    20   0 8477516 522048 194192 S   9.1   3.2  11:39.54 gnome-shell
18853 ubuntu    20   0 22536    5516   3356 R   9.1   0.0   0:00.02 top
   1 root      20   0 27056    17864 11784 S   0.0   0.1   0:04.24 systemd
   2 root      20   0      0      0      0 S   0.0   0.0   0:00.04 kthreadd
   3 root      20   0      0      0      0 S   0.0   0.0   0:00.00 pool_workqueue_release
   4 root      0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
   5 root      0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
   6 root      0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
   7 root      0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-slub_flushwq
   8 root      0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-netns
  10 root      0 -20      0      0      0 I   0.0   0.0   0:00.61 kworker/0:0H-kblockd
  13 root      0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-mm_percpu_wq
  14 root      20   0      0      0      0 S   0.0   0.0   0:00.06 ksoftirqd/0
  15 root      20   0      0      0      0 I   0.0   0.0   0:13.33 rcu_preempt
  16 root      20   0      0      0      0 S   0.0   0.0   0:00.00 rcu_exp_par_gp_kthread_worker/0
  17 root      20   0      0      0      0 S   0.0   0.0   0:00.40 rcu_exp_gp_kthread_worker
  18 root      rt   0      0      0      0 S   0.0   0.0   0:00.12 migration/0
  19 root      20   0      0      0      0 S   0.0   0.0   0:00.00 kprobe-optimizer
  20 root     -51   0      0      0      0 S   0.0   0.0   0:00.00 idle_inject/0
  21 root      20   0      0      0      0 S   0.0   0.0   0:00.00 cpuhp/0
  22 root      20   0      0      0      0 S   0.0   0.0   0:00.00 cpuhp/1
  23 root     -51   0      0      0      0 S   0.0   0.0   0:00.00 idle_inject/1
  24 root      rt   0      0      0      0 S   0.0   0.0   0:00.11 migration/1
  25 root      20   0      0      0      0 S   0.0   0.0   0:00.07 ksoftirqd/1
  27 root      0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/1:0H-kblockd
  28 root      20   0      0      0      0 S   0.0   0.0   0:00.00 cpuhp/2
  29 root     -51   0      0      0      0 S   0.0   0.0   0:00.00 idle_inject/2
  30 root      rt   0      0      0      0 S   0.0   0.0   0:00.11 migration/2
  31 root      20   0      0      0      0 S   0.0   0.0   0:00.07 ksoftirqd/2
  33 root      0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/2:0H-kblockd
  34 root      20   0      0      0      0 S   0.0   0.0   0:00.00 cpuhp/3
  35 root     -51   0      0      0      0 S   0.0   0.0   0:00.00 idle_inject/3

```

El comando top (Table of Processes) es un monitor de rendimiento del sistema interactivo y en tiempo real, estándar en sistemas operativos tipo Unix. Su función principal es proporcionar una vista dinámica y centralizada de la actividad del sistema, mostrando el uso global de los recursos de hardware (CPU, memoria) junto con una lista detallada de los procesos que consumen más recursos en el sistema.

• Htop

```
ubuntu@ubuntu: ~ -- htop

0[          0.0%] 3[          0.0%] 5[          0.0%] 8[|||||||100.0%]
1[          0.0%] 4[          0.0%] 6[          0.0%] 9[          0.0%]
2[          0.0%]

Mem[|||||||3.42G/15.6G] Tasks: 148, 845 thr, 215 kthr; 1 running
Swp[          0K/0K] Load average: 0.27 0.46 0.54
Uptime: 00:55:38

Main I/O
PID USER PRI NI VIRT RES SHR S CPU% MEM% TIME+ Command
18882 ubuntu 20 0 21740 6064 3556 R 41.7 0.0 0:00.04 htop
1 root 20 0 27056 17864 11784 S 0.0 0.1 0:04.25 /sbin/init autoinstall
977 root 19 -1 50968 10908 17016 S 0.0 0.1 0:00.62 /usr/lib/systemd/systemd-journald
1031 systemd-oo 20 0 16808 7748 6700 S 0.0 0.0 0:00.55 /usr/lib/systemd/systemd-oomd
1032 systemd-re 20 0 22920 15548 12572 S 0.0 0.1 0:02.03 /usr/lib/systemd/systemd-resolved
1037 root 20 0 41572 14964 8960 S 0.0 0.1 0:00.53 /usr/lib/systemd/systemd-udev
1906 avahi 20 0 6748 4724 4332 S 0.0 0.0 0:00.17 avahi-daemon: running [ubuntu.local]
1907 root 20 0 2888 2100 1980 S 0.0 0.0 0:00.04 /bin/sh /usr/lib/systemd/scripts/chronyd-starter.sh -n -F 1
1908 messagebus 20 0 12608 9228 4768 S 0.0 0.1 0:02.43 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --syslog
1912 root 20 0 58472 30224 14960 S 0.0 0.2 0:01.29 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-triggers
1915 polkitd 20 0 316M 12360 8540 S 0.0 0.1 0:00.31 /usr/lib/polkit-1/polkitd --no-debug --log-level=notice
1920 root 20 0 2348M 53464 29768 S 0.0 0.3 0:01.32 /usr/lib/swapd/swapd
1923 root 20 0 313M 7804 7012 S 0.0 0.0 0:00.29 /usr/libexec/accounts-daemon
1927 root 20 0 10752 2908 2668 S 0.0 0.0 0:00.04 /usr/sbin/cron -f -P
1930 root 20 0 312M 7760 6980 S 0.0 0.0 0:00.04 /usr/libexec/switcheroo-control
1933 root 20 0 10604 5692 8248 S 0.0 0.1 0:00.47 /usr/lib/systemd/systemd-logind
1937 root 20 0 163M 7104 5668 S 0.0 0.0 0:00.02 /usr/sbin/zed -F
1948 avahi 20 0 6560 1692 1388 S 0.0 0.0 0:00.00 avahi-daemon: chroot helper
1959 root 20 0 163M 7104 0 S 0.0 0.0 0:00.00 /usr/sbin/zed -F
1960 root 20 0 163M 7104 0 S 0.0 0.0 0:00.00 /usr/sbin/zed -F
1967 root 20 0 312M 7760 0 S 0.0 0.0 0:00.00 /usr/libexec/switcheroo-control
1968 root 20 0 312M 7760 0 S 0.0 0.0 0:00.00 /usr/libexec/switcheroo-control
1970 root 20 0 313M 7804 0 S 0.0 0.0 0:00.00 /usr/libexec/accounts-daemon
1971 root 20 0 313M 7804 0 S 0.0 0.0 0:00.09 /usr/libexec/accounts-daemon
1976 polkitd 20 0 316M 12360 0 S 0.0 0.1 0:00.05 /usr/lib/polkit-1/polkitd --no-debug --log-level=notice
1992 root 20 0 531M 15444 12652 S 0.0 0.1 0:00.17 /usr/libexec/udisks2/udisksd
2010 syslog 20 0 215M 5828 4604 S 0.0 0.0 0:00.14 /usr/sbin/rsyslogd -n -iNONE
2022 polkitd 20 0 316M 12360 0 S 0.0 0.1 0:00.00 /usr/lib/polkit-1/polkitd --no-debug --log-level=notice
F1 Help F2 Setup F3 Search F4 Filter F5 Tree F6 Sort By F7 Nice F8 Vmice F9 Kill F10 Quit
```

```
ubuntu@ubuntu1:~$ sudo apt install htop
Instalando:
 htop

Paquetes sugeridos:
 lm-sensors

Resumen:
 Actualizando: 0, Instalando 1, Eliminando: 0, no actualizando: 141
 Tamaño de la descarga: 177 kB
 Espacio necesario: 451 kB / 42,7 GB disponible

Des:1 http://pe.archive.ubuntu.com/ubuntu resolute/main amd64 htop amd64 3.4.1-5build2 [177 kB]
Descargados 177 kB en 1s (299 kB/s)
Seleccionando el paquete htop previamente no seleccionado.
(Leyendo la base de datos ... 154028 ficheros o directorios instalados actualmente.)
Preparando para desempaquetar .../htop_3.4.1-5build2_amd64.deb ...
Desempaquetando htop (3.4.1-5build2) ...
Configurando htop (3.4.1-5build2) ...
Procesando disparadores para desktop-file-utils (0.28-1build1) ...
Procesando disparadores para hicolor-icon-theme (0.18-2build1) ...
Procesando disparadores para gnome-menus (3.38.1-1ubuntu1) ...
Procesando disparadores para man-db (2.13.1-1build1) ...
ubuntu@ubuntu1:~$ htop
```

El comando htop es un monitor de procesos e instrumentación del sistema interactivo, diseñado como una alternativa moderna y mejorada del comando tradicional top. A diferencia de su predecesor, htop proporciona una interfaz gráfica basada en texto (TUI) a todo color, soporte para el uso del ratón, desplazamiento vertical y horizontal de la lista de procesos, y una representación visual mucho más intuitiva del estado del hardware. Previamente, se debe instalar mediante el comando `sudo apt install htop`.

• Nice

```
ubuntu@ubuntu: ~  
ubuntu@ubuntu:~$ sudo nice -n -10 sudo -u $USER DBUS_SESSION_BUS_ADDRESS=$DBUS_SESSION_BUS_ADDRESS snap-store &  
[1] 19340  
ubuntu@ubuntu:~$ libEGL warning: failed to get driver name for fd -1  
  
libEGL warning: MESA-LOADER: failed to retrieve device information  
  
libEGL warning: failed to get driver name for fd -1  
  
MESA: error: ZINK: failed to choose pdev  
libEGL warning: egl: failed to create dri2 screen  
Gdk-Message: 19:34:05.036: Unable to load from the cursor theme  
INFO snap-store: Logging to /home/ubuntu/snap/snap-store/1390/.local/share/snap-store/snap-store.log  
INFO appstream: Unsupported locale: C. Defaulting to "en".  
ERROR error_observer: Provider ratingsModelProvider:RatingsModelProvider#bc149(null) failed  
GrpcError: gRPC Error (code: 14, codeName: UNAVAILABLE, message: Error connecting: SocketException: Connection refused (OS Error: Connection refused,  
errno = 111), address = ratings.ubuntu.com, port = 44810, details: null, rawResponse: null, trailers: {})  
ERROR error_observer: Provider ratingsModelProvider:RatingsModelProvider#adb44(null) failed  
GrpcError: gRPC Error (code: 14, codeName: UNAVAILABLE, message: Error connecting: SocketException: Connection refused (OS Error: Connection refused,  
errno = 111), address = ratings.ubuntu.com, port = 44810, details: null, rawResponse: null, trailers: {})  
ERROR error_observer: Provider ratingsModelProvider:RatingsModelProvider#f5237(null) failed  
GrpcError: gRPC Error (code: 14, codeName: UNAVAILABLE, message: Error connecting: SocketException: Connection refused (OS Error: Connection refused,  
errno = 111), address = ratings.ubuntu.com, port = 44810, details: null, rawResponse: null, trailers: {})  
ERROR error_observer: Provider ratingsModelProvider:RatingsModelProvider#af443(null) failed  
GrpcError: gRPC Error (code: 14, codeName: UNAVAILABLE, message: Error connecting: SocketException: Connection refused (OS Error: Connection refused,  
errno = 111), address = ratings.ubuntu.com, port = 44810, details: null, rawResponse: null, trailers: {})  
ERROR error_observer: Provider ratingsModelProvider:RatingsModelProvider#898c9(null) failed  
GrpcError: gRPC Error (code: 14, codeName: UNAVAILABLE, message: Error connecting: SocketException: Connection refused (OS Error: Connection refused,  
errno = 111), address = ratings.ubuntu.com, port = 44810, details: null, rawResponse: null, trailers: {})  
ERROR error_observer: Provider ratingsModelProvider:RatingsModelProvider#a3f5f(null) failed  
GrpcError: gRPC Error (code: 14, codeName: UNAVAILABLE, message: Error connecting: SocketException: Connection refused (OS Error: Connection refused,  
errno = 111), address = ratings.ubuntu.com, port = 44810, details: null, rawResponse: null, trailers: {})  
ERROR error_observer: Provider ratingsModelProvider:RatingsModelProvider#77d0(null) failed  
GrpcError: gRPC Error (code: 14, codeName: UNAVAILABLE, message: Error connecting: SocketException: Connection refused (OS Error: Connection refused,  
errno = 111), address = ratings.ubuntu.com, port = 44810, details: null, rawResponse: null, trailers: {})  
ERROR error_observer: Provider ratingsModelProvider:RatingsModelProvider#43d3f(null) failed  
GrpcError: gRPC Error (code: 14, codeName: UNAVAILABLE, message: Error connecting: SocketException: Connection refused (OS Error: Connection refused,  
errno = 111), address = ratings.ubuntu.com, port = 44810, details: null, rawResponse: null, trailers: {})  
ERROR error_observer: Provider ratingsModelProvider:RatingsModelProvider#1a446(null) failed  
GrpcError: gRPC Error (code: 14, codeName: UNAVAILABLE, message: Error connecting: SocketException: Connection refused (OS Error: Connection refused,  
errno = 111), address = ratings.ubuntu.com, port = 44810, details: null, rawResponse: null, trailers: {})  
  
INFO snap-store: Logging to /home/ubuntu/snap/snap-store/1390/.local/share/snap-store/snap-store.log  
INFO appstream: Unsupported locale: C. Defaulting to "en".  
ERROR error_observer: Provider ratingsModelProvider:RatingsModelProvider#bc149(null) failed  
GrpcError: gRPC Error (code: 14, codeName: UNAVAILABLE, message: Error connecting: SocketException: Connection refused (OS Error: Connection refused,  
errno = 111), address = ratings.ubuntu.com, port = 44810, details: null, rawResponse: null, trailers: {})  
ERROR error_observer: Provider ratingsModelProvider:RatingsModelProvider#adb44(null) failed  
GrpcError: gRPC Error (code: 14, codeName: UNAVAILABLE, message: Error connecting: SocketException: Connection refused (OS Error: Connection refused,  
errno = 111), address = ratings.ubuntu.com, port = 44810, details: null, rawResponse: null, trailers: {})  
ERROR error_observer: Provider ratingsModelProvider:RatingsModelProvider#f5237(null) failed  
GrpcError: gRPC Error (code: 14, codeName: UNAVAILABLE, message: Error connecting: SocketException: Connection refused (OS Error: Connection refused,  
errno = 111), address = ratings.ubuntu.com, port = 44810, details: null, rawResponse: null, trailers: {})  
ERROR error_observer: Provider ratingsModelProvider:RatingsModelProvider#af443(null) failed  
GrpcError: gRPC Error (code: 14, codeName: UNAVAILABLE, message: Error connecting: SocketException: Connection refused (OS Error: Connection refused,  
errno = 111), address = ratings.ubuntu.com, port = 44810, details: null, rawResponse: null, trailers: {})  
ERROR error_observer: Provider ratingsModelProvider:RatingsModelProvider#898c9(null) failed  
GrpcError: gRPC Error (code: 14, codeName: UNAVAILABLE, message: Error connecting: SocketException: Connection refused (OS Error: Connection refused,  
errno = 111), address = ratings.ubuntu.com, port = 44810, details: null, rawResponse: null, trailers: {})  
ERROR error_observer: Provider ratingsModelProvider:RatingsModelProvider#a3f5f(null) failed  
GrpcError: gRPC Error (code: 14, codeName: UNAVAILABLE, message: Error connecting: SocketException: Connection refused (OS Error: Connection refused,  
errno = 111), address = ratings.ubuntu.com, port = 44810, details: null, rawResponse: null, trailers: {})  
ERROR error_observer: Provider ratingsModelProvider:RatingsModelProvider#77d0(null) failed  
GrpcError: gRPC Error (code: 14, codeName: UNAVAILABLE, message: Error connecting: SocketException: Connection refused (OS Error: Connection refused,  
errno = 111), address = ratings.ubuntu.com, port = 44810, details: null, rawResponse: null, trailers: {})  
ERROR error_observer: Provider ratingsModelProvider:RatingsModelProvider#43d3f(null) failed  
GrpcError: gRPC Error (code: 14, codeName: UNAVAILABLE, message: Error connecting: SocketException: Connection refused (OS Error: Connection refused,  
errno = 111), address = ratings.ubuntu.com, port = 44810, details: null, rawResponse: null, trailers: {})  
ERROR error_observer: Provider ratingsModelProvider:RatingsModelProvider#1a446(null) failed  
GrpcError: gRPC Error (code: 14, codeName: UNAVAILABLE, message: Error connecting: SocketException: Connection refused (OS Error: Connection refused,  
errno = 111), address = ratings.ubuntu.com, port = 44810, details: null, rawResponse: null, trailers: {})  
ERROR error_observer: Provider ratingsModelProvider:RatingsModelProvider#56f15(null) failed  
GrpcError: gRPC Error (code: 14, codeName: UNAVAILABLE, message: Error connecting: SocketException: Connection refused (OS Error: Connection refused,  
errno = 111), address = ratings.ubuntu.com, port = 44810, details: null, rawResponse: null, trailers: {})  
ERROR error_observer: Provider ratingsModelProvider:RatingsModelProvider#d8c27(null) failed  
GrpcError: gRPC Error (code: 14, codeName: UNAVAILABLE, message: Error connecting: SocketException: Connection refused (OS Error: Connection refused,  
errno = 111), address = ratings.ubuntu.com, port = 44810, details: null, rawResponse: null, trailers: {})  
ERROR error_observer: Provider ratingsModelProvider:RatingsModelProvider#af6ba(null) failed  
GrpcError: gRPC Error (code: 14, codeName: UNAVAILABLE, message: Error connecting: SocketException: Connection refused (OS Error: Connection refused,  
errno = 111), address = ratings.ubuntu.com, port = 44810, details: null, rawResponse: null, trailers: {})
```

El comando nice es una utilidad nativa del sistema operativo que permite modificar la prioridad de asignación de tiempo de CPU de un proceso en el momento de su lanzamiento. En Linux, el programador del núcleo decide qué procesos reciben tiempo de procesamiento basado, entre otros factores, en un valor numérico denominado valor de cortesía. El comando nice altera este valor para evitar que procesos secundarios saturen el procesador.

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